Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 October 2026
Team ID	SWTID-2026-9874
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture

The proposed solution is implemented within the ServiceNow platform. A Service Desk Agent uses a web browser to access the Incident form or list. The ServiceNow client layer applies UI Policies and Client Scripts to control field behavior, validate user actions, and provide feedback. Valid Incident data is stored in the ServiceNow Incident record/database.

Architecture Diagram – ServiceNow Incident Control

Service Desk Agent	→	**Web Browser / ServiceNow UI**	→	**ServiceNow Platform**
User input	→	Incident Form / Incident List	→	UI Policies & Client Scripts
		↓		
		Business Rules / Validation Logic		
		↓		
		Incident Record / Data Store		
		Saved / Updated Incident		

Infrastructure demarcation: The ServiceNow application, platform services, and Incident data store are cloud-hosted within the ServiceNow environment. The user's browser is the client-side access point.

External interfaces: No third-party API is required for the current project scope. The solution operates using native ServiceNow Incident functionality, UI Policies, and Client Scripts.

Machine learning interface: No machine learning model is required or used in this project.

Table-1: Components & Technologies

S.No	Component	Description / Technology

1	User Interface	ServiceNow Incident form and Incident list accessed through a web browser. Technology: ServiceNow web UI / HTML, CSS and JavaScript-based client interface.
2	Application Logic-1	High Impact UI Policy. Controls Assignment Group mandatory behavior and Urgency read-only behavior based on Impact.
3	Application Logic-2	onChange Client Script. Automatically sets Urgency to High when Impact changes to High.
4	Application Logic-3	onSubmit Client Script and onCellEdit Client Script. onSubmit validates Assigned To before saving; onCellEdit restricts State changes from list editing.
5	Database	ServiceNow Incident table / platform data store. Stores Incident fields such as Impact, Urgency, Assignment Group, Assigned To and State.
6	Cloud Database	ServiceNow-managed platform data storage. No separate external cloud database is required for this project.
7	File Storage	No separate application file-storage service is required for the current Incident-control solution.
8	External API-1	None required. The project uses native ServiceNow platform capabilities.
9	External API-2	None required in the current scope.
10	Machine Learning Model	Not applicable. No ML model is used.
11	Infrastructure (Server / Cloud)	Cloud-hosted ServiceNow platform accessed through a web browser. No separate application server is deployed by the project team.

Table-2: Application Characteristics

S.No	Characteristics	Description / Technology
1	Open-Source Frameworks	No external open-source application framework is required for the current project. The implementation uses ServiceNow's native UI Policy and Client Script mechanisms with JavaScript.
2	Security Implementations	Access is provided through ServiceNow user authentication and role-based access controls. Client-side validation restricts the configured Incident actions, including direct State list editing. Server-side platform access controls remain part of the ServiceNow environment.
3	Scalable Architecture	The solution is lightweight and platform-based. UI Policies and Client Scripts are applied at the Incident interface and can be extended with additional rules without introducing a separate application server.
4	Availability	The solution runs within the ServiceNow platform and is accessed through the web interface. No separate local server is required for the project.
5	Performance	Client-side field checks execute on relevant form events such as onChange, onSubmit and onCellEdit. The scripts are small and eventspecific, minimizing unnecessary processing during normal Incident operations.

Technology Stack Summary

- Platform: ServiceNow
- Interface: ServiceNow Incident Form and List
- Client-side scripting: JavaScript / ServiceNow Client Scripts
- UI control: ServiceNow UI Policies and UI Policy Actions
- Data storage: ServiceNow Incident table / platform-managed storage
- Deployment: ServiceNow cloud platform
- External APIs: None required
- Machine Learning: Not applicable

Reference note: The supplied template requires the architecture to show application processes, infrastructure demarcation, external interfaces, data storage, and ML interfaces where applicable. Those categories are explicitly addressed above.